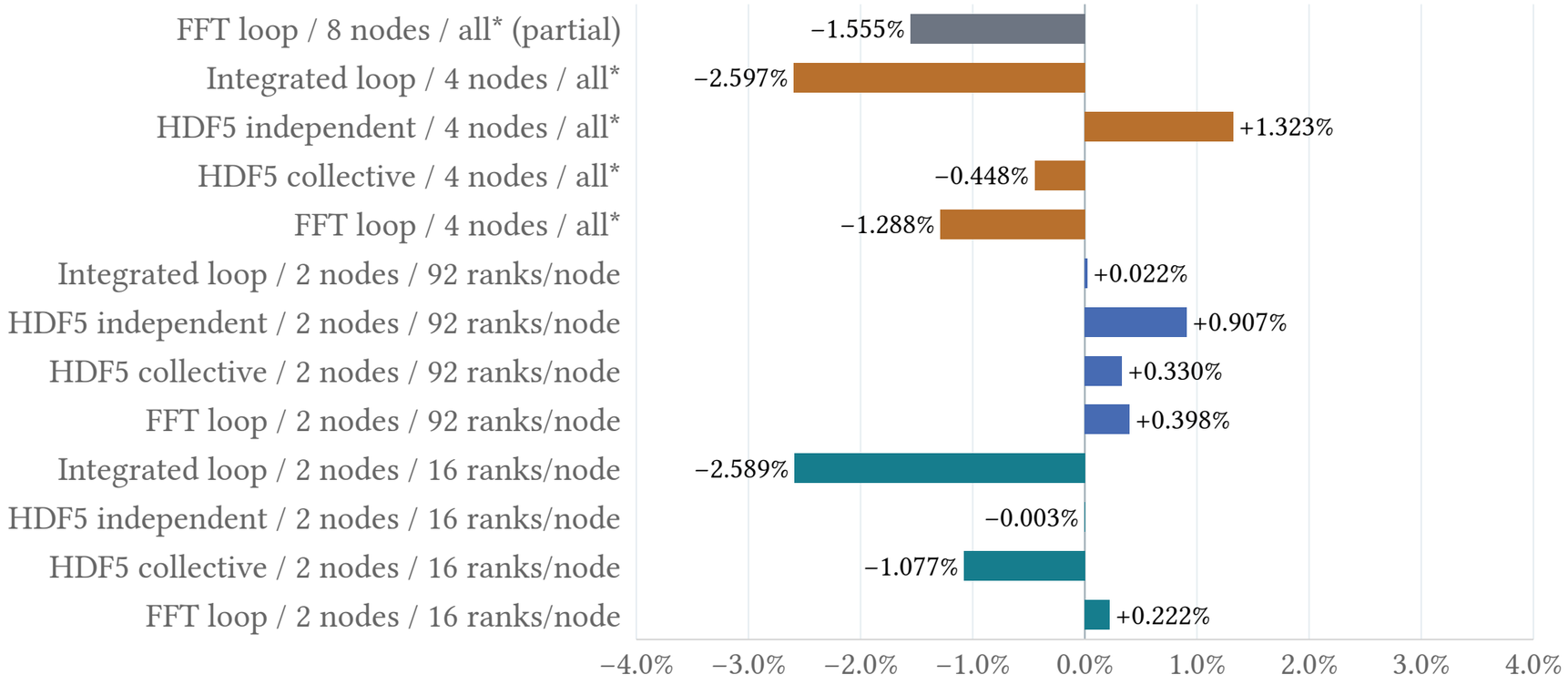
Compiler hardening: results so far

Completed point estimates: −2.60% to +1.32%

Full = candidate hardening profile. Reference = ordinary build. Negative = faster. Positive = slower.



Coverage

12 complete conditions
1 partial
3 not measured

Missing: 8-node HDF5 + integrated

Sample counts

10 pairs per complete case
(20 process runs)

Partial: 4/10 pairs

One pair = full + reference

Provisional screenshot transcription, 16 Oct 2026. Error bars unavailable. Original build verification pending.

Takeaway: no consistent slowdown is apparent across these completed cases. Coverage and definitions follow on slide 2.

Small point estimates do not prove zero overhead. *Exact 4-/8-node core counts are not visible in the photos.

Test design, time and remaining coverage

Dense MPI phase only. Synthetic library workflows. Earlier serial/package screens are separate.
Full and reference change the caller plus the complete trial-owned library stack.

Workloads constructed

FFT loop	3D forward/inverse complex FFTs. Plans reused.
HDF5 collective	Collective writes, flush/close and read-back.
HDF5 independent	Independent writes, flush/close and read-back.
Integrated loop	Repeated FFTs with periodic checkpoints.

How the comparison works

Maximum-rank loop time excludes startup, planning and final read-back.
Balanced process pairs. Work and rank layout stay fixed within each pair.
One rank per physical core. Slurm/Open MPI uses the saved UCX fabric policy.
Exact sizes, settings and build treatment await original-record verification.

Time and sign

Change = 100 x (full/reference - 1).
Positive: slower. Negative: faster.

Example only: a 12-hour job

At +1%: 7 min 12 sec longer.
At −1%: 7 min 12 sec shorter.

Assumes a sustained whole-job effect
and comparable workload behavior.
More hours add seconds, not percentage.

Actual run duration is unavailable from the photos.

Follow-up before a deployment decision

Audit flags and raw pairs. Test a fixed ordinary caller at 20 nodes, then vary FFT grids and sustained loop duration.
Vary checkpoint size, cadence and read/write patterns. Add OSU and repeat in independent allocations.
Current coverage does not establish whole-application, cold-cache or durable-storage cost.